

CU Online — Chandigarh University

MCA Final Year Project — Milestone Report

MILESTONE 4 REPORT

Build Visualization Dashboard

Implementing the 9-visual interactive Power BI dashboard with DAX measures and cross-filtering

Project Title: Multi-Cloud Cost Comparison and Optimization Analytics Dashboard

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Program: Master of Computer Applications (MCA) | CU Online

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Milestone 4: Build Visualization Dashboard

This Milestone is the central implementation milestone of the project — the phase in which the data model from Milestone 3 was transformed into the interactive, user-facing Power BI dashboard. This milestone encompassed the definition of all four DAX calculated measures, the implementation of all nine dashboard visuals on the primary report page, the configuration of interactive slicers, and the creation of the provider detail drill-through page.

4.1 DAX Measures Implemented

Four DAX measures were defined in the data model before building any visuals, as the visuals depend on these measures for their calculated values.

Measure Name	DAX Formula	Purpose	Visual(s) Using It
Total Cloud Cost	SUM('Raw Dataset'[Cost (\$M)])	Total infrastructure cost in current filter context	KPI Card 1, Bar Chart (V2), Donut Chart (V4)
GCP Savings vs AWS	CALCULATE([...],Provider=AWS) - CALCULATE([...],Provider=GCP)	Full-period AWS–GCP savings (ignores slicer via CALCULATE)	KPI Card 3
Avg Cost Per User	AVERAGE('Raw Dataset'[Cost per User (\$K)])	Average efficiency metric in current filter context	KPI Card 4, Line Chart (V3)
Total Users	SUM('Raw Dataset'[New Users (K)])	Total user count in current filter context	KPI Card 2, Area Chart (V5)

Table M4.1: DAX Measures Implemented in Milestone 4

4.2 Dashboard Visuals Implemented

Visual No.	Type	Data Source	Fields Used	Key Configuration
V1	KPI Cards (×4)	Raw Dataset	Total Cloud Cost, Total Users, GCP Savings, Avg Cost/User	28pt bold; GCP Savings in green (#34A853)
V2	Clustered Bar Chart	Raw Dataset	Financial Year (Y), Cost \$M (X), Provider (Legend)	Brand colours; data labels ON; title descriptive
V3	Line Chart	Raw Dataset	Financial Year (X), Cost/User \$K (Y), Provider (Legend)	Markers ON; Y-axis starts at 0
V4	Donut Chart	Raw Dataset	Cost \$M (Values), Provider (Legend)	Inner radius 50%; percentage labels ON
V5	Stacked Area Chart	Raw Dataset	Financial Year (X), New Users K (Y), Provider (Legend)	20% layer transparency

Visual No.	Type	Data Source	Fields Used	Key Configuration
V6	Drill-Down Column Chart	Quarterly Breakdown	FY/Quarter hierarchy (X), Cost \$M (Y), Provider (Legend)	Drill mode ON; hierarchy auto-created
V7	Slicers (×2)	Raw Dataset	Provider (buttons); Financial Year (dropdown)	Provider tiles use brand colours
V8	Table Visual	Free Tier & Storage	Provider/Service, Free Storage, Duration, Notes	Conditional format: GCP 15 GB = green
V9	Multi-Row Card / Table	Recommendations	Provider, Strategy, Headline, Est. Savings	Row conditional formatting by provider

Table M4.2: Dashboard Visuals Implemented — Configuration Summary

4.3 Provider Detail Drill-Through Page Created

A second report page (Provider Detail) was created as a drill-through target. The page contains four provider-specific trend charts: Quarterly Cost Line Chart, Storage Capacity Bar Chart, Network Speed Line Chart, and Cost-per-User Line Chart. The drill-through field (Provider) was added to the page's Drill through well in the Visualizations panel, enabling right-click drill-through navigation from any provider data point on the main page.

4.4 Cross-Filtering and Interactivity Verified

All nine visuals on the primary page were verified to respond correctly to both slicer selections: the Provider slicer (AWS, Azure, GCP buttons) and the Financial Year slicer (FY2021–FY2025 dropdown). The GCP Savings vs AWS KPI card was verified to correctly ignore the Provider slicer (returning \$100.4M regardless of which provider is selected), confirming that the CALCULATE function's filter-context override is working correctly.

4.5 Dashboard Formatting Standards Applied

- Canvas size: 1920 × 1080 pixels (Full HD). Page background: #F8F9FA.
- Provider brand colours applied consistently: AWS = #FF9900, Azure = #008AD7, GCP = #34A853.
- All chart titles: Segoe UI 12pt Bold, colour #1F4E79.
- Horizontal grid lines only, colour #E8E8E8, 1px weight.
- Visual container background: white with light grey border.